Industrial Internship Report

on

"Forecasting of Smart City Traffic Patterns"

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project was developed to predict traffic at different city junctions using machine learning. It uses historical traffic data to identify traffic patterns. Features like hour, day, and previous traffic values are created. Linear Regression, Decision Tree, and Random Forest models are compared. Random Forest performed best and was used for final traffic forecasting. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

Summary of the whole 6 weeks' work.

About need of relevant Internship in career development.

Brief about Your project/problem statement.

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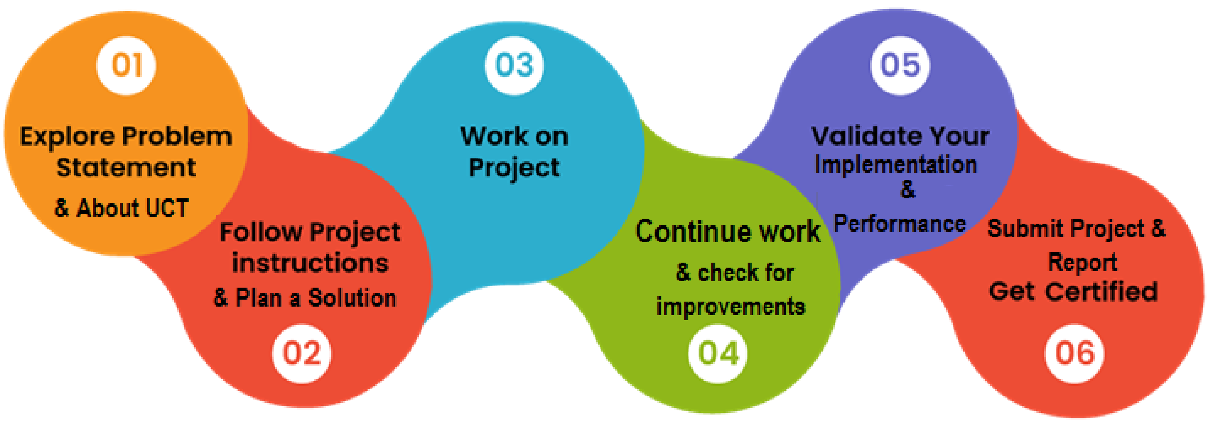


Your Learnings and overall experience.

Thank to all (with names), who have helped you directly or indirectly.

Your message to your juniors and peers.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

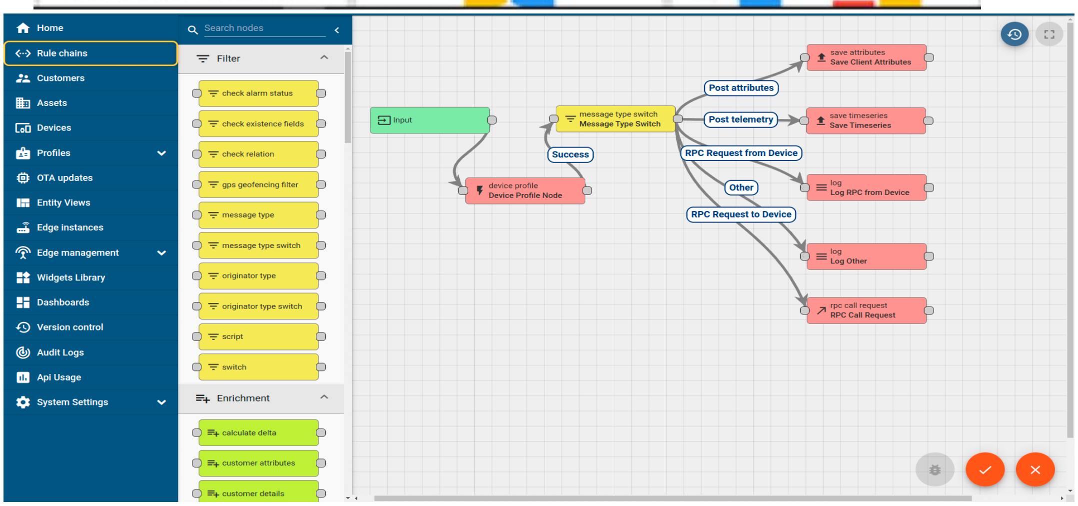
i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to
• Build Your own dashboard
• Analytics and Reporting
• Alert and Notification

- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



ii. Smart Factory Platform (FACTORY WATCH)

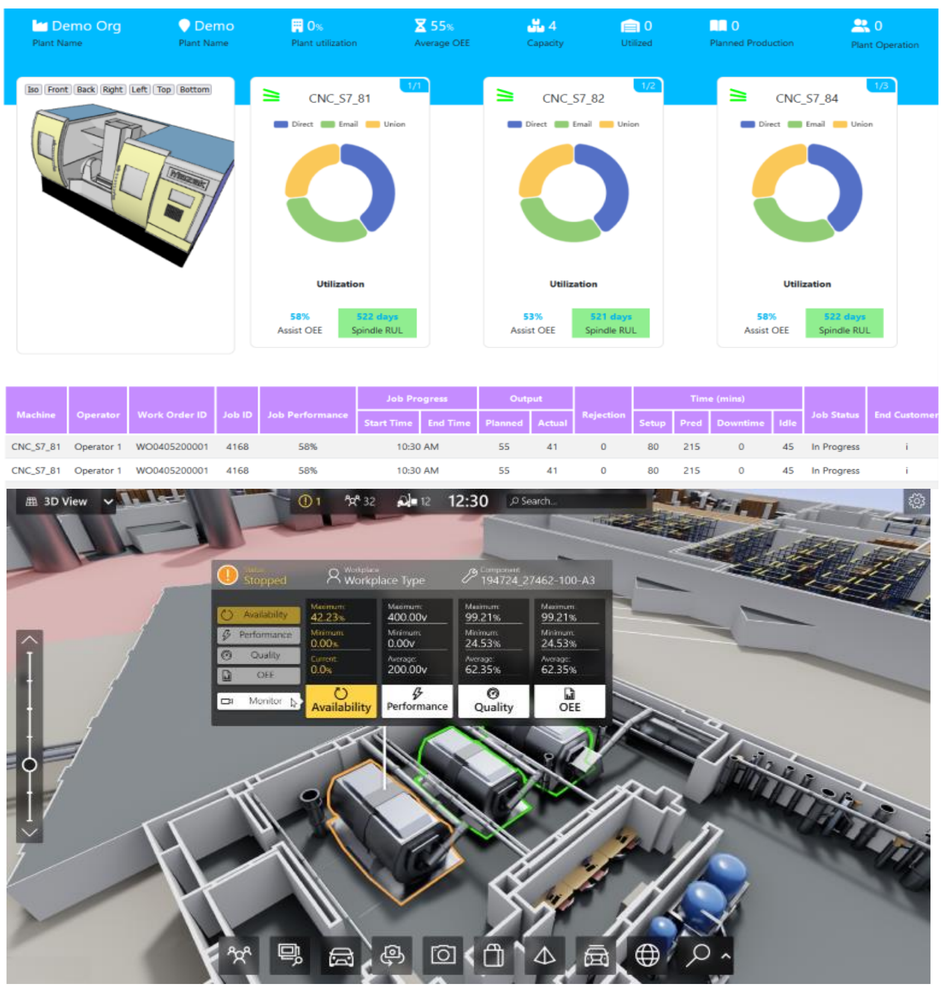
Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring

- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

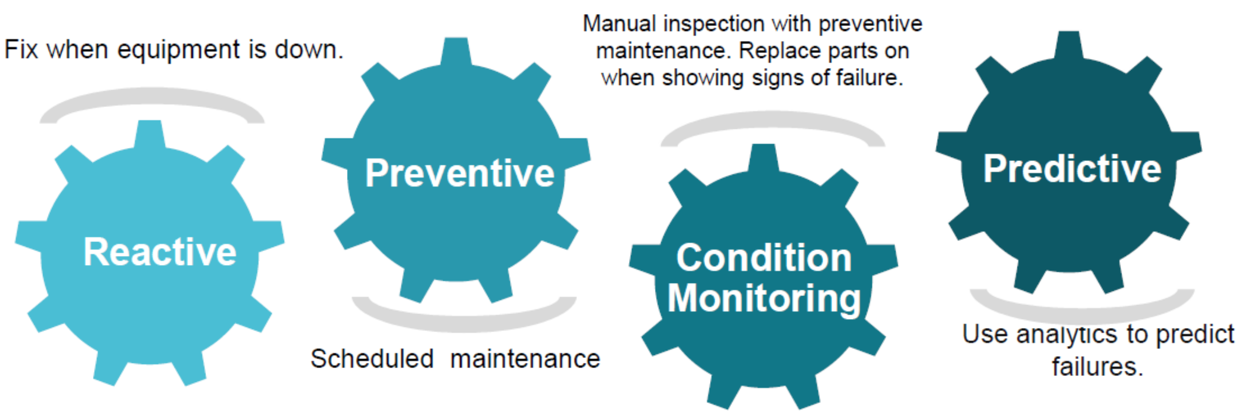


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

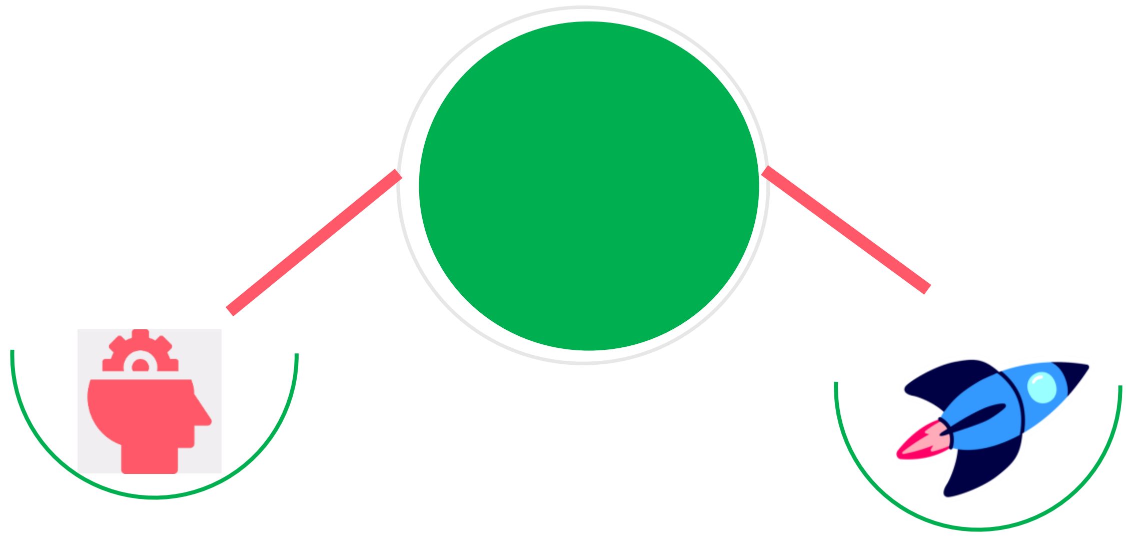
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

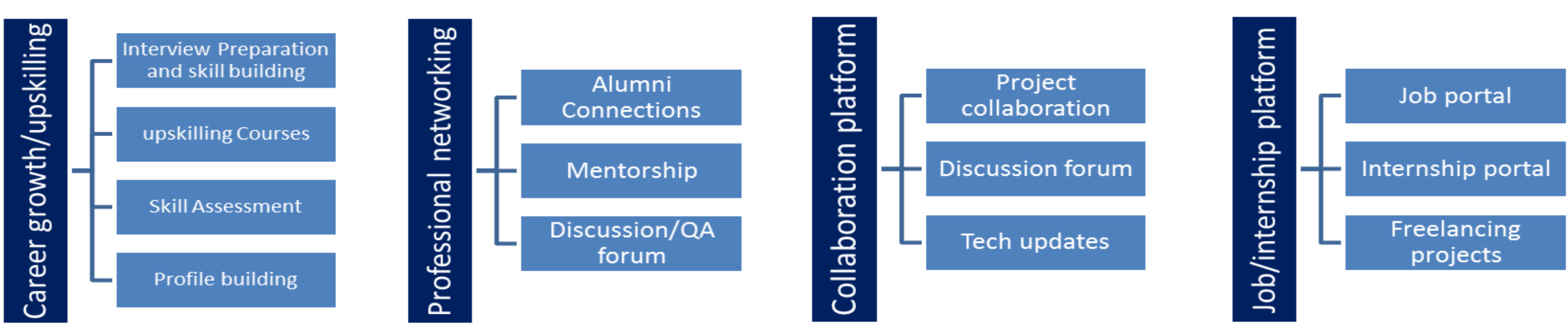
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] **Kaggle – Smart City Traffic Patterns Dataset**
Used as the main dataset for traffic forecasting.
https://github.com/ChandniRana/upskill_campus/blob/main/datasets_8494_11879_test_BdBKkAj.csv

[2] **Python Documentation**
Used for implementing the project and data processing.
https://www.python.org/doc/

[3]**Pandas Documentation**
Used for data loading, cleaning, and manipulation.
https://pandas.pydata.org/docs/

[4]**Scikit-learn Documentation**
Used for implementing Linear Regression, Decision Tree, Random Forest, and evaluation metrics.
https://scikit-learn.org/stable/

[5]**Matplotlib Documentation**
Used for creating graphs and visualizing traffic patterns.
https://matplotlib.org/stable/

2.6 Glossary

Term	Meaning
EDA	Exploratory Data Analysis
ML	Machine Learning
MAE	Mean Absolute Error
RMSE	Root Mean Squared Error
R^2	Coefficient of Determination
Lag1	Previous hour's traffic count
Lag24	Traffic count from 24 hours earlier
ID	Record identifier
CSV	Comma-Separated Values

3 Problem Statement

The project addresses the problem of forecasting hourly traffic volume at multiple city junctions using historical traffic observations. Traffic patterns vary with time of day, day of week, junction and recent traffic conditions. A forecasting system is needed to estimate future vehicle counts for a period where the actual target values are not available.

Input data

Training data: 48,120 labelled hourly records.

Test data: 11,808 hourly records without the Vehicles target.

Four monitored junctions are represented by the Junction field.

DateTime provides the chronological information required for temporal features.

Problem requirements

Prepare the data without breaking chronological order.

Capture short-term and daily traffic dependencies.

Compare multiple regression models using meaningful validation metrics.

Generate forecasts for the complete test period.

Expected output

The expected output is a forecast for the Vehicles value for every test record, saved in submission.csv along with the corresponding ID. The best trained model is also saved as traffic_prediction_model.pkl.

A major challenge is that the test period has no ground-truth Vehicles values. Therefore, the final forecasting stage uses autoregressive lag updates, where each predicted hour becomes available for later lag-based predictions.

4 Existing and Proposed solution

Existing approach / limitation

A simple approach to traffic prediction can use only fixed calendar variables or a single regression model. Such an approach may not capture the strong relationship between consecutive traffic observations. A single decision tree can also overfit local patterns and may provide weaker generalization.

Proposed solution

The implemented solution uses an end-to-end time-series-aware machine learning workflow. It combines calendar features with recent-history features and compares three regression algorithms under a chronological validation strategy.

Data cleaning and chronological checks.

EDA of distributions, hourly/weekday patterns and feature correlations.

Time features: Hour, Day, DayOfWeek, Month, WeekNumber and Quarter.

Lag1, Lag24 and 24-hour RollingMean features computed per junction.

Time-based validation using the last 20% of each junction's history.

Model comparison using MAE, RMSE and R^2.

Random Forest selected as the final model because it achieved the best validation results.

Final model refitted on the complete labelled training data.

Autoregressive forecasting used for the unlabeled test period.

4.1 Code submission : https://github.com/Mamatha-10/upskillcampus/commit/969b18260a91e16444b01cbb1481d0175b438385

4.2 Report submission

5 Proposed Design/ Model

The project follows a sequential data science workflow. Historical traffic data is processed first, useful temporal and historical features are created, candidate regression models are trained and evaluated, the best model is selected and then used for forecasting the unlabeled test period.

5.1 High Level Diagram

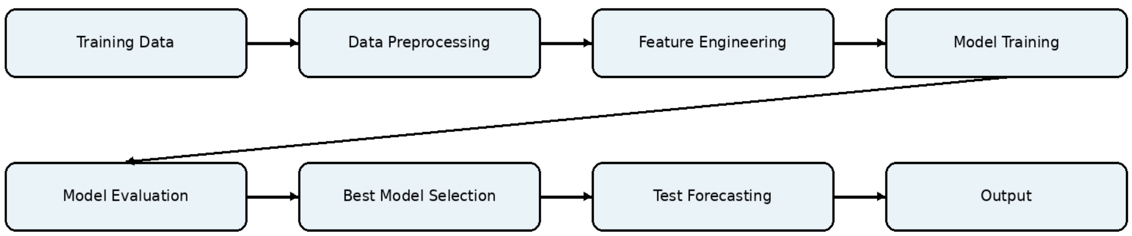


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Interfaces

The project is primarily a Python-based batch/command-line implementation rather than a graphical web application. Its practical interfaces are the input CSV files, Python training script, saved model and generated forecast CSV.

- Input interface: data/train.csv and data/test.csv.
- Processing interface: train_models.py.
- Model interface: traffic_prediction_model.pkl.
- Output interface: data/submission.csv and report/metric files.

6 Performance Test

The performance of the Smart City Traffic Forecasting system was evaluated based on **prediction accuracy and model performance**. The main constraint was to obtain accurate traffic predictions while selecting the model with the lowest prediction error.

Identified Constraints

- **Prediction Accuracy:** The model should predict traffic volume close to the actual values.
- **Prediction Error:** MAE and RMSE should be as low as possible.
- **Model Performance:** R^2 should be as high as possible.
- **Processing:** The system should be able to process historical traffic data and generate forecasts.

How the Constraints Were Handled

Historical traffic data was preprocessed and time-based features such as **Hour, Day, Day of Week, Lag1, Lag24, and Rolling Mean** were created. Three machine learning models—**Linear Regression, Decision Tree, and Random Forest**—were trained and compared using MAE, RMSE, and R^2.

6.1 Test Plan/ Test Cases

Test Case	**Test Activity**	**Expected Result**	**Status**
TC-01	Load training and test datasets	Files load with expected columns	Pass
TC-02	Validate data quality	No missing/duplicate issues in analyzed data	Pass
TC-03	Create temporal features	Expected time features generated	Pass
TC-04	Create lag/rolling features	Lag1, Lag24 and RollingMean generated per junction	Pass
TC-05	Train candidate models	All three models train successfully	Pass
TC-06	Evaluate models	MAE, RMSE and R^2 produced	Pass

TC-07	Select best model	Random Forest selected	Pass
TC-08	Forecast test period	11,808 test predictions generated	Pass
TC-09	Save model/output	Model and submission CSV created	Pass

6.2 Test Procedure

The training history was split chronologically, with the final 20% of each junction used as validation data. Each candidate model was trained on earlier observations and evaluated on the held-out period. After comparison, Random Forest was refit using all labeled training data. The model was then used to generate the test-period forecasts.

6.3 Performance Outcome

Model	MAE	RMSE	R^2 Score
Random Forest	3.3649	5.3766	0.9605
Linear Regression	3.7256	5.6238	0.9568
Decision Tree	3.9928	6.6345	0.9399

Random Forest produced the strongest validation result, with the lowest MAE and RMSE and the highest R^2. The final forecasting process generated 11,808 test predictions.

- Feature importance analysis showed that Lag1 was the dominant feature at approximately 96.8%, followed by Hour, Lag24 and RollingMean. This indicates that recent traffic history is the strongest signal used by the final model. Analysis outputs: metrics.csv, feature_importances.png and EDA plots.

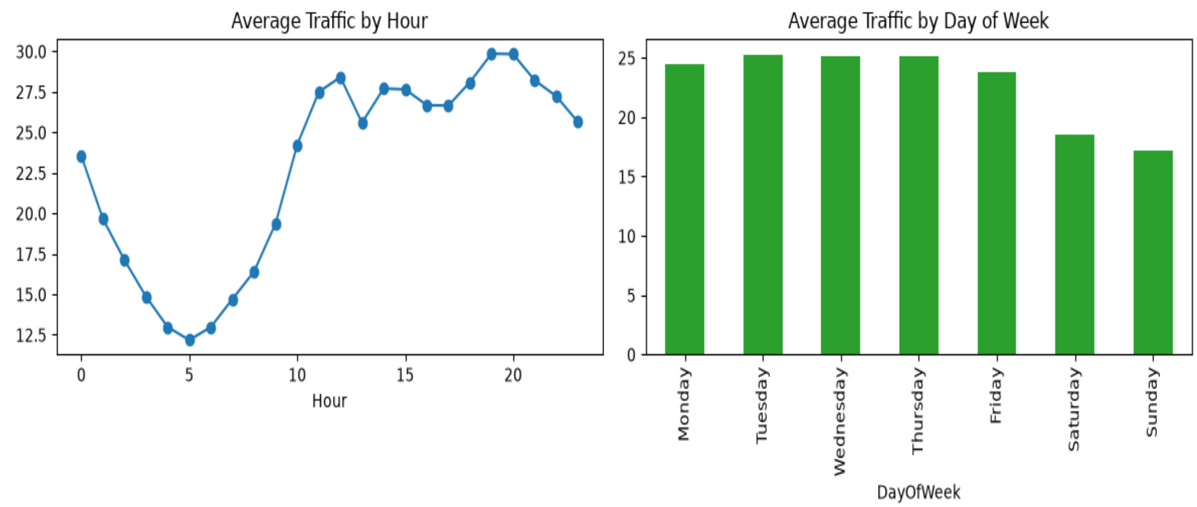


Figure 3: Hourly and weekday traffic pattern from the project analysis.

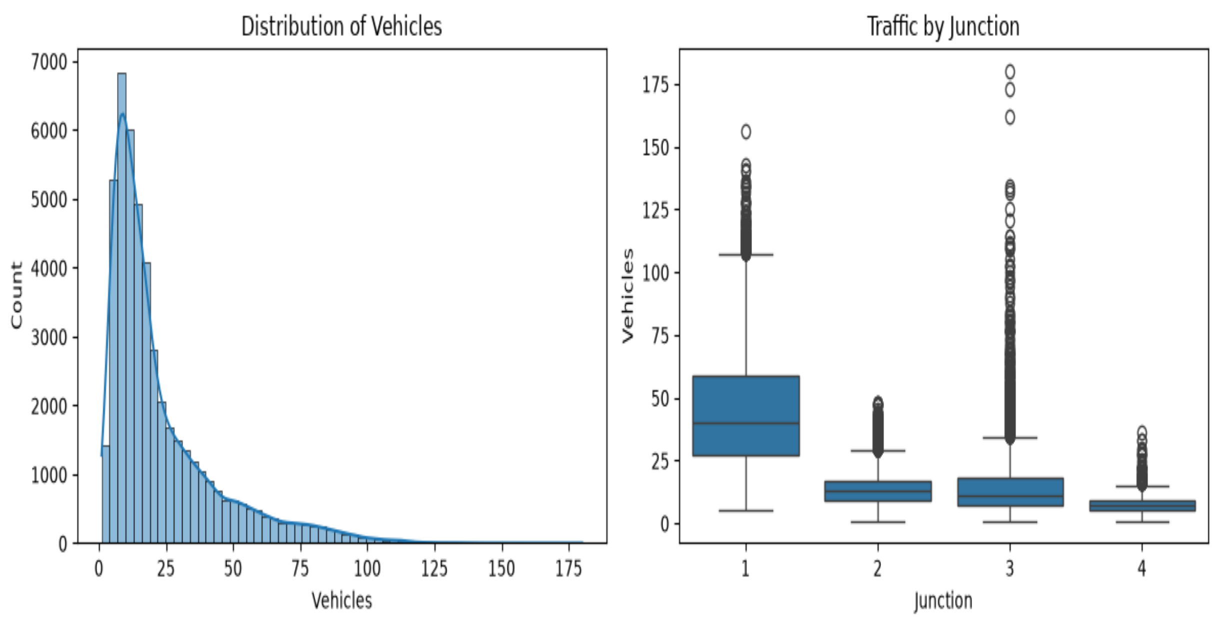


Figure 4: Traffic distribution / junction-wise analysis from the project.

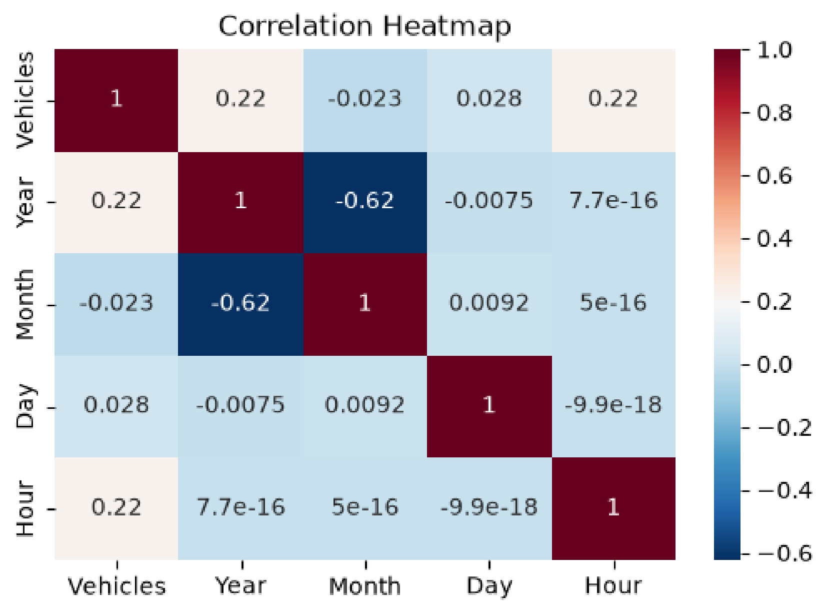


Figure 5: Correlation analysis generated by the project.

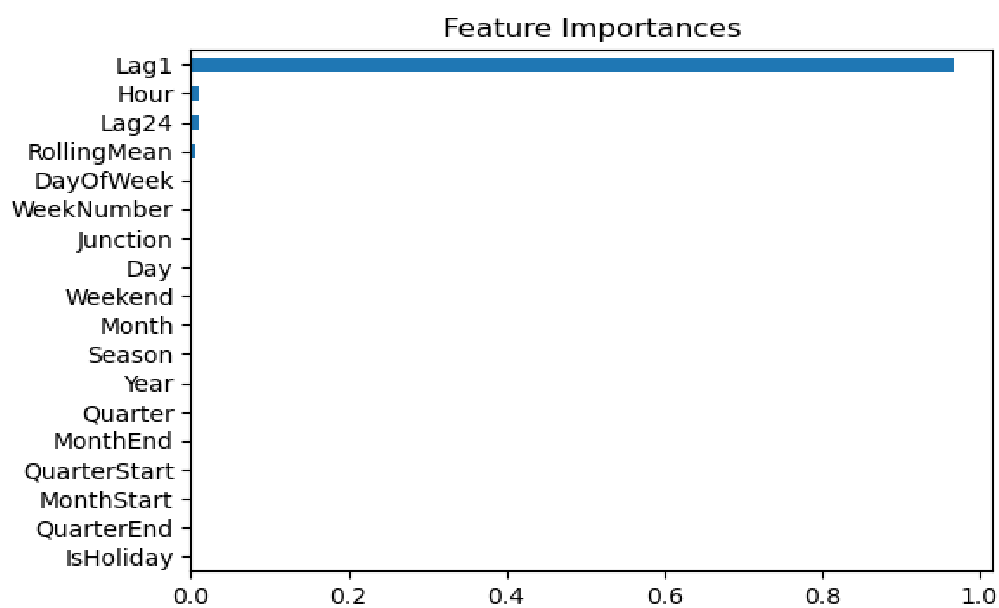


Figure 6: Feature importance of the final Random Forest model.

7 My learnings

- Learned how to work with real-world time-series traffic data using Pandas.
- Learned how DateTime information can be converted into useful machine learning features.
- Understood the use of Lag1, Lag24 and RollingMean features for traffic forecasting.
- Gained practical experience with Linear Regression, Decision Tree and Random Forest regression.
- Learned how to compare models using MAE, RMSE and R^2.
- Understood why chronological validation is suitable for time-dependent data.
- Learned how autoregressive forecasting can be used when future target values are unavailable.
- Improved debugging, dataset-path handling and end-to-end testing skills.
- Learned how to save and reuse a trained machine learning model using Joblib.

8 Future work scope

- Use advanced sequence models such as LSTM or GRU to capture longer traffic dependencies.
- Include weather information and a more complete holiday/event calendar to improve forecasting.
- Integrate live traffic data for near-real-time traffic prediction.
- Develop an interactive interface for users to view traffic forecasts.
- Deploy the trained model through a cloud-based API.
- Study methods to reduce error accumulation during long autoregressive forecasting.

These are future enhancement ideas and were not treated as completed parts of the current internship project.